BIOLOGY BOARD QUESTION PAPER : MARCH 2023

Time: 3 Hrs. Max. Marks: 70

General Instructions:

The question paper is divided into four sections.

(1) Section A: Q. No. 1 contains Ten multiple choice type of questions carrying one mark each.

(i) For each multiple choice type of question, it is mandatory to write the correct answer along with its alphabet, e.g., (A) ……. / (B) ……. / (C) ……. / (D) ……. etc. No marks shall be given if ONLY the correct answer or alphabet of the correct answer is written.

(ii) In case of MCQ, evaluation will be done for the first attempt only.

Q. No. 2 contains Eight very short answer type questions carrying one mark each.

(2) Section B: Q. No. 3 to 14 are short answer type questions carrying two marks each. (Attempt any Eight)

(3) Section C: Q. No. 15 to 26 are short answer type questions carrying three marks each. (Attempt any Eight)

(4) Section D: Q. No. 27 to 31 are long answer type of questions carrying four marks each. (Attempt any Three)

(5) Begin the answer of each section on a new page.

SECTION – A

Q.1 Select and write the correct answer for the following multiple choice type of questions: [10]

i. Histones are rich in _______.

(A) Lysine and Arginine (B) Leucine and Methionine

(C) Serine and Leucine (D) Phenyl alanine and Lysine

ii. How many mitotic divisions take place during the formation of a female gametophyte from a functional megaspore?

(A) One (B) Two (C) Three (D) Four

iii. Which of the following is the only gaseous plant growth regulator?

(A) ABA (B) Cytokinin (C) Ethylene (D) Gibberellin

iv. The pH of nutrient medium for plant tissue culture is in the range of _______.

(A) 2 to 4.2 (B) 5 to 5.8 (C) 7 to 7.5 (D) 8 to 9.5

v. Rivet Popper Hypothesis is an analogy to explain the significance of _______.
(A) Biodiversity (B) natality (C) sex-ratio (D) age distribution ratio

vi. Which of the following group shows ZW-ZZ type of sex determination?
(A) Pigeon, Parrot, Sparrow (B) Parrot, Bat, Fowl
(C) Bat, Fowl, Crow (D) Sparrow, Fowl, Cat

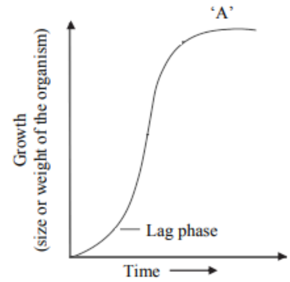
vii. In Hamburger's phenomenon, _______.
(A) Cl– diffuse into WBCs (B) Cl– diffuse into RBCs
(C) Na+ diffuse into RBCs (D) Na+ diffuse into WBCs

viii. Calcium and Phosphate ions are balanced between blood and other tissues by ______.
(A) Thymosin and Parathormone (B) Calcitonin and Somatostatin
(C) Collip's hormone and Calcitonin (D) Calcitonin and Thymosin

ix. Identify the INCORRECT statement.
(A) In a flaccid cell, T.P. is zero (B) In a turgid cell, DPD is zero
(C) In a fully turgid cell, TP = OP (D) Water potential of pure water is negative

x. Which of the following is a hormone releasing contraceptive?
(A) Cu-T (B) Cu-7 (C) Multiload-375 (D) LNG-20

Q.2 Answer the following questions: [8]
i. Which disease is caused by HPV?
ii. Which device is used to clean both dust and gases from polluted air?
iii. Mention the name of sterile animal produced by intergeneric hybridisation.
iv. Give the name of first transgenic plant.
v. A child has low BMR, delayed puberty and mental retardation. Identify the disease.
vi. Identify 'A' in the given graph of population growth.



vii. Complete the following box with reference to symptoms of mineral deficiency:

Abscission	Pre mature fall of flowers, fruits and leaves
______________________	Appearance of green and non green patches on leaves

viii. Give an example of plant having both kidney and dumb-bell shaped guard cells in stomata.

SECTION – B

(Attempt any EIGHT) [16]

Q.3 Define the terms: a. Gross Primary Productivity b. Net Primary Productivity

Q.4 Draw a neat diagram of thyroid gland and label thyroid follicle, follicular cells and blood capillaries.

Q.5 i. Give reason – ABA is also known as antitranspirant.

ii. Explain the role of chlorophyllase enzyme in banana.

Q.6 Complete the chart and re-write:

Disorders/Diseases	– Recombinant Proteins
?	– Erythropoietin
Asthma	– ?
?	– Tissue plasminogen activator
Emphysema	– ?

Q.7 i. Define – Imbibition

ii. Explain how imbibition helps root hairs in adsorption of water.

Q.8 Draw a neat diagram of conducting system of human heart and label AV node, Bundle of His and Purkinje fibres.

Q.9 Distinguish between heterochromatin and euchromatin with reference to staining property and activity.

Q.10 Complete the following chart regarding energy flow and re-write:

?	Herbivores
Primary Producer	?
?	Man, Lion
Secondary consumer	?

Q.11 i. What is biofortification?

ii. Mention one example each of fortification with reference to: a. Amino acid content

b. Vitamin-C content

Q.12 Differentiate between X-chromosome and Y-chromosome with reference to:

i. Length of non-homologous regions ii. Type as per position of centromere

Q.13 Define: i. Genetic drift ii. Homologous organs

Q.14 i. What is ex-situ conservation? ii. Mention any two places where ex-situ conservation is undertaken.

SECTION – C

(Attempt any EIGHT) [24]

Q.15 i. Define – Incomplete dominance.

ii. If a red flowered Mirabilis jalapa plant is crossed with a white flowered plant, what will be the phenotypic ratio in F2 generation? Show it by a chart.

Q.16 i. Differentiate between sympathetic and parasympathetic nervous system with reference to: a. Pre and post ganglionic nerve fibres b. Effect on heart beat

ii. Give reason – All spinal nerves are of mixed type.

Q.17 i. Draw a suitable diagram of replication of eukaryotic DNA and label any three parts.

ii. How many amino acids will be there in the polypeptide chain formed on the following mRNA? 5' GCCACAUGGAGAUGACGACAAAUUUACUAGAAAA 3'

Q.18 Describe the steps in breathing.

Q.19 i. What is spermatogenesis?

ii. Draw a neat and labelled diagram of spermatogenesis.

Q.20 i. What is a connecting link?

ii. Which fossil animal is considered as the connecting link between reptiles and birds? Give one character of each class.

Q.21 Complete the following chart and re-write:

1. _____ – Plasmodium and Man
2. _____ – Leopard and Lion
3. _____ – Clown fish and Sea-anemone

Q.22 i. What is composition of bio-gas?

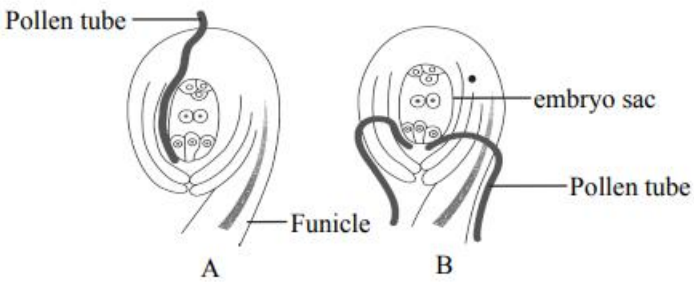
ii. Mention any four benefits of bio-gas.

Q.23 i. Give reason – Water acts as thermal buffer.

ii. Draw a neat diagram of root hair and label mitochondria, nucleus and vacuole.

Q.24 Explain three main functions of free antibodies produced by B-lymphocytes.

Q.25 i. Following are the diagrams of entry of pollen tube into ovule. Identify the type A and B.



ii. Give any four points of significance of double fertilization.

Q.26 i. Name the hormone responsible for apical dominance.
ii. Suggest any plant hormone to remove broad-leaved weeds from jowar plantation.
iii. Mention any two applications of cytokinin.

SECTION – D

(Attempt any THREE) [12]

Q.27 i. What is blood pressure?
ii. Name the instrument used to measure blood pressure.
iii. Differentiate between an artery and a vein with reference to lumen and thickness of wall.

Q.28 i. Describe any three adaptations in anemophilous flowers. Mention one example.
ii. Describe any three adaptations in hydrophilous flowers. Mention one example.

Q.29 i. What is polymerase chain reaction (PCR)?
ii. Describe three steps involved in mechanism of PCR.

Q.30 i. Give any four significances of fertilization in humans.
ii. Mention the names of any two organs each derived from ectoderm and mesoderm.

Q.31 i. Give any two functions of cerebellum.
ii. Write names of any four motor cranial nerves with their serial number.
iii. Which hormones stimulate liver for glycogenesis and glucogenolysis?